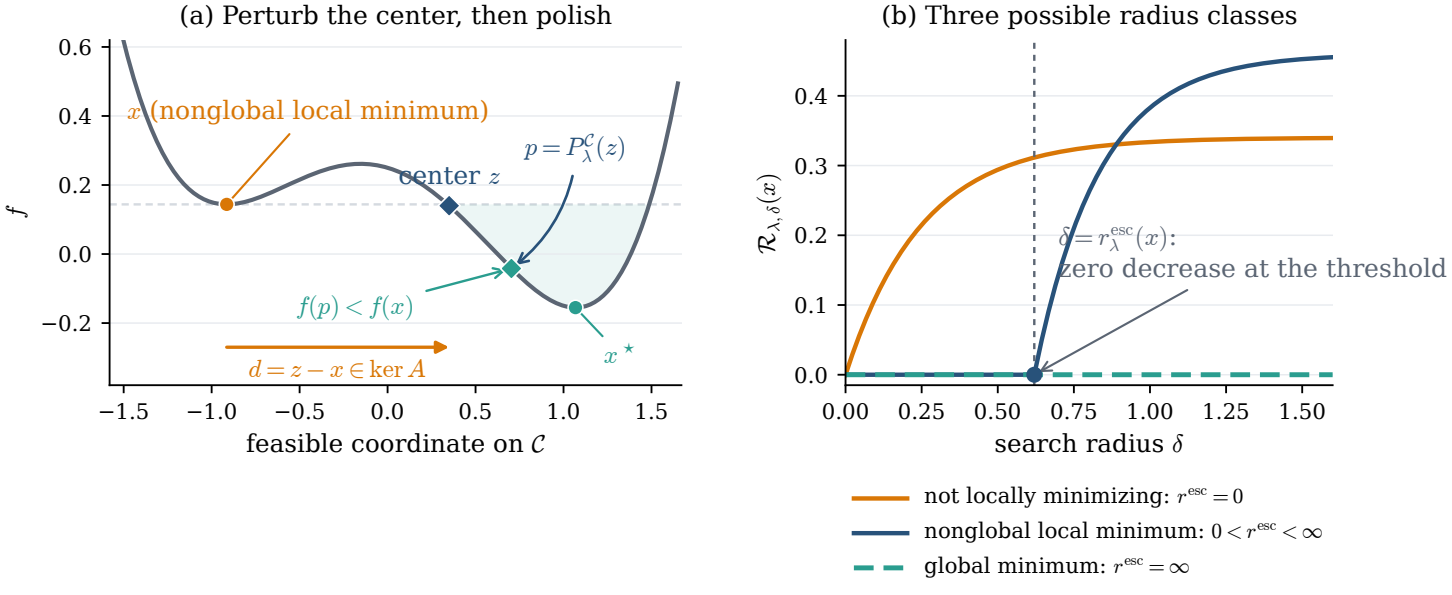
(a) Perturb the center, then polish
x (nonglobal local minimum)
center z
$p = P_\lambda^{\mathcal{C}}(z)$
$f(p) < f(x)$
$x^\star$
$d = z - x \in \ker A$
f
feasible coordinate on $\mathcal{C}$
(b) Three possible radius classes
$\mathcal{R}_{\lambda,\delta}(x)$
$\delta = r_\lambda^{\mathrm{esc}}(x)$:
zero decrease at the threshold
search radius δ
not locally minimizing: $r^{\mathrm{esc}} = 0$
nonglobal local minimum: $0 < r^{\mathrm{esc}} < \infty$
global minimum: $r^{\mathrm{esc}} = \infty$